

AI Base³ Solutions — NSF SBIR Phase I Proposal: Revision Matrix & Running Change Log

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Project Workspace: SINE~WaiV Technical Repository

Target Solicitation: NSF SBIR Phase I (NSF 26-510)

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1. Executive Running Change Log

The following table tracks the specific modifications executed during the Session 1 Claim Audit to eliminate overclaims, bifurcate architecture modes, and establish rigorous falsification criteria.

Issue ID	Target Section	Original Technical Risk / Problem	Executed Modification (Clean Text)	Rigor Justification
ISSUE 1	Part II-B, C4 — Competitive Advantage	Unproven claims of absolute superiority over incumbent MCSA solutions in VFD environments.	"Phase I will test whether the design addresses documented MCSA limitations in VFD environments."	Removes false maturity; frames the project as a scientifically rigorous test rather than a pre-validated conclusion.
ISSUE 2	Part I — Broader Impacts	Conflation of software-only deployment (Mode A) and next-generation autonomous hardware (Mode B).	"Phase I establishes Mode A (software-only architecture). Phase II extends the platform to Mode B autonomous sensing, targeting ATEX-rated hazardous areas where vibration sensor	Eliminates reviewer confusion regarding Phase I deliverables versus the longer-term commercial hardware roadmap.

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			installation is prohibitive."	
ISSUE 3	Part III, Section 2.1	Potential domain boundary overlap between current data and vibration data.	<i>[No Action Required]</i> Proposal correctly segregates Paderborn KAt (current-domain validation) from NASA IMS (vibration-reference validation).	Preserves clean segregation of empirical validation boundaries.
ISSUE 4	Part IV, Section 8.2.1 — Statistical Grounding	Missing explicit falsification criteria; defined success but omitted boundaries for performance failure.	"If $K-S D \leq 0.15$ ($p > 0.05$), the system logs insufficient spectral separation, and the Phase II technical roadmap will pivot to alternative mixed-radix sub-windowing architectures."	Demonstrates project governance discipline by defining clear statistical failure modes for reviewers.
ISSUE 5	Part II-B, C3 — Revenue Model	Overconfident tone projecting guaranteed Phase II/III SaaS contracts and commercial OEM integration.	"Anticipated commercialization pathway: beta SaaS platform deployment with select industrial partners and potential OEM co-development opportunities."	Aligns language with market realism; emphasizes validation over premature commercial certainty.
ISSUE 6	Part III, Section 3.3 — VFD Robustness	Ambiguity regarding the relative baseline	" $SNR \geq 9$ dB across all VFD artifact injection	Establishes an unambiguous, absolute

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		of the "< 3 dB degradation" metric.	conditions. This absolute threshold ensures fault detection feasibility independent of Aim 1 outcomes."	performance floor aligned with standard industrial practice.
ISSUE 7	Part III, Section 6.3 — Edge Compute Feasibility	Ambiguity over whether the 40 mW constraint is a Phase I gate or a Phase II optimization target.	"The 40 mW target enables the Phase II energy-scavenging Spectral Guardian Dot™. If consumption exceeds 40 mW during Phase I evaluation, it will be classified as a Phase II optimization vector; Mode A (software-only) deployment remains viable regardless."	Prevents a narrow edge-hardware specification miss from invalidating core Phase I algorithmic milestones.
ISSUE 8	Multiple Sections (Objective, Part II, Section 2)	Loose use of the formal control-theory term "observability" introducing semantic slippage.	Standardized terminology to: "fault-relevant feature detection" or "soft mechanical-state inference" across all non-control-theory contexts.	Signals domain-specific precision to specialist academic and industrial reviewers.
ISSUE 9	Part III, Section 5 — Current Evidence Stack	Lack of explicit, quantified "Proceed Gates"	Integrated explicit "Proceed to Next Aim"	Unambiguously links experimental

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		mapping individual components to technical Aims.	rows requiring defined statistical bounds (e.g., $K-S D \geq 0.3$ & $KL \Delta D_{kl} > 0.5$).	outcomes to project advancement milestones.

Note: Line spacing for this ledger and associated document components is maintained at a professional 1.25 layout with Arial-compatible clear heading hierarchies. No dozenal terminology or unauthorized IP designations have been utilized in these texts.

2. Executed Technical Text Replacements

Part I — Broader Impacts [ISSUE 2]

Revised Clean Text:

The broader commercial and societal impacts of the proposed framework scale across two primary deployment vectors. Phase I establishes Mode A (software-only architecture), allowing immediate integration into existing legacy variable frequency drive (VFD) data capture loops without requiring infrastructure modifications. Phase II extends the platform to Mode B autonomous sensing, targeting ATEX-rated hazardous areas where traditional vibration sensor installation is physically or economically prohibitive. By bypassing the requirement for per-bearing physical sensor installation in hazardous environments, this dual-stage optimization minimizes initial capital expenditure while expanding advanced predictive maintenance monitoring to historically underserved small-to-medium manufacturing environments.

Part II-B, C3 — Revenue Model [ISSUE 5]

Revised Clean Text:

The anticipated commercialization pathway focuses on a phased market-entry strategy designed to minimize capital drag. Initial deployment targets beta SaaS platform deployment with select industrial partners to validate algorithm stability under diverse operating regimes. Following successful operational field-testing, the organization will pursue potential OEM co-development opportunities, embedding the validated SINE~WaiV sub-windowing algorithms directly into next-generation industrial drive controllers. This leverages existing hardware distribution channels without accelerating premature direct sales infrastructure costs during early validation phases.

Part II-B, C4 — Competitive Advantage [ISSUE 1 & ISSUE 8]

Revised Clean Text:

Legacy motor current signature analysis (MCSA) systems frequently encounter steep performance degradation in environments dominated by non-linear loads and dynamic switching

frequencies. Phase I will test whether the design addresses documented MCSA limitations in VFD environments. By deploying a localized 65,536-point Cooley-Tukey primary Fast Fourier Transform (FFT) combined with a targeted 1,728-point mixed-radix sub-window topology, the SINE~WaiV architecture isolates fault-relevant feature detection from background electrical noise without assuming baseline stationarity. This approach transforms standard current data into a high-fidelity diagnostic stream, optimizing soft mechanical-state inference at the edge.

Part III, Section 3.3 — VFD Robustness [ISSUE 6]

Revised Clean Text:

To establish firm engineering baselines for variable frequency drive noise rejection, technical success is evaluated against fixed empirical thresholds rather than comparative baselines. The system must maintain an $\text{SNR} \geq 9 \text{ dB}$ across all VFD artifact injection conditions. This absolute threshold ensures fault detection feasibility independent of Aim 1 outcomes, locking down an uncorrupted processing band outside the primary VFD switching noise envelopes (specifically targeting the empirical exclusion bands located at 170–185 Hz and 258–260 Hz).

Part III, Section 5 — Current Evidence Stack & Technical Governance [ISSUE 9]

To ensure strict structural discipline during Phase I execution, progression between core research aims is gated by the quantified evidence matrix detailed below.

Evidence Component / Source	Technical Target Phase	Quantified Proceed Gate	Falsification/Deflection Threshold
Paderborn KAt Dataset (Current-Domain)	Aim 1 Execution	$\text{K-S } D \geq 0.30; p < 0.01$	$\text{K-S } D \leq 0.15$; Fail to reject null hypothesis.
NASA IMS Reference Dataset (Vibration Baseline)	Aim 2 Cross-Validation	$\text{KL } \Delta D_{\text{kl}} > 0.50$ cross-domain match	$\text{KL } \Delta D_{\text{kl}} \leq 0.20$; Divergent signal envelope.
VFD Noise Injection Loop	Aim 3 Robustness Check	Absolute $\text{SNR} \geq 9 \text{ dB}$ across active switching states	Absolute $\text{SNR} < 6 \text{ dB}$; Trigger VFD kill-zone filter adjustment.

Part III, Section 6.3 — Edge Compute Feasibility [ISSUE 7]

Revised Clean Text:

Algorithmic execution footprints are evaluated against hard hardware computational ceilings. The 40 mW target enables the Phase II energy-scavenging Spectral Guardian Dot™. If consumption exceeds 40 mW during Phase I bench evaluation, it will be classified as a Phase II optimization vector rather than an immediate project showstopper; Mode A (software-only) deployment remains viable regardless, as its execution footprint is absorbed entirely by existing host-drive microprocessor overhead. This boundaries the physical power constraints to prevent

early hardware specification variances from halting core algorithmic validation milestones.

Part IV, Section 8.2.1 — Statistical Grounding [ISSUE 4]

Revised Clean Text:

Statistical verification of the feature separation engine relies on the Kolmogorov-Smirnov (K-S) test statistic applied to the isolated defect harmonics. A successful verification outcome is defined as achieving a K-S $D \geq 0.3$ with an associated significance of $p < 0.01$. Conversely, explicit falsification criteria are established to govern performance failure: if K-S $D \leq 0.15$ ($p > 0.05$), the system logs insufficient spectral separation, indicating that background VFD transient harmonics have successfully masked the low-amplitude mechanical fault signatures. In the event this falsification threshold is met, Phase II technical roadmaps will pivot to alternative mixed-radix sub-windowing architectures or tighter empirical notch filtering strategies, preventing ongoing resource commitment to an unseparated spectral profile.

End of Audit Application Document. All institutional boundaries verified.